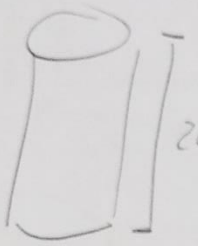
LISTA 2

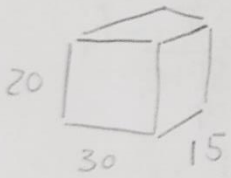
① 

$P = e \cdot g \cdot H$

$P = 1000 \cdot 10 \cdot 20 = 200000\ Pa$

$P = 200\ kPa$

② $\rho = \frac{m}{v} = \frac{6 \cdot 10^{24}}{20 \cdot 10^{19}} = 3 \cdot 10^{4}\ kg/m^3$

③ 

$d = 7,6 \Rightarrow \rho = 7600\ kg/m^3$

$g = 10\ m/s^2$

a) $m = 0,2 \cdot 0,3 \cdot 0,15 \cdot 7600 = 68,4\ kg$

b) $P_{máx} = \frac{68,4 \cdot 10}{0,2 \cdot 0,15} = 22800\ Pa$

$22,8\ kPa$

$P_{mín} = \frac{68,4 \cdot 10}{0,2 \cdot 0,3} = 11200\ Pa$

$11,2\ kPa$

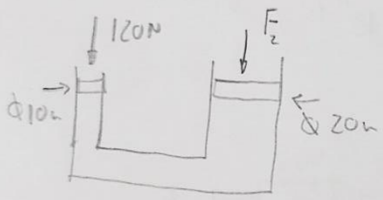
④ $\Delta P = \rho \cdot g \cdot \Delta H$

$\Delta P = 13600 \cdot 10 \cdot (0,25 - 0,15)$

$\Delta P = 13600\ Pa = 13,6\ kPa$

$P = \Delta P + P_{atm} = 13,6 + 100 = \boxed{113,6\ kPa}$

⑤

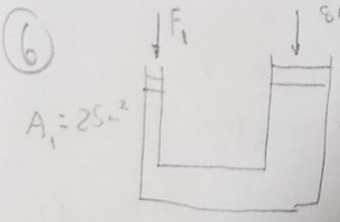


$F_1/A_1 = F_2/A_2$

$F_2 = \frac{A_2 \cdot F_1}{A_1} = \frac{\frac{\pi \cdot D_2^2}{4}}{\frac{\pi \cdot D_1^2}{4}} \cdot F_1$

$F_2 = \frac{D_2^2}{D_1^2} \cdot F_1 = \frac{20^2}{10^2} \cdot 120 = \boxed{480N}$

⑥

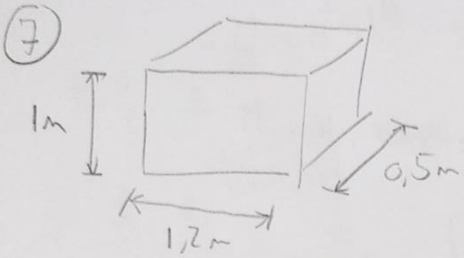


$\frac{F_1}{A_1} = \frac{F_2}{A_2}$

$\frac{F_1}{25} = \frac{8000}{2000}$

$\boxed{F_1 = 100N}$

(7)



$P_{CV} = 540\ kgf$

a) $P = \frac{P_{CV}}{A_B} = \frac{540}{0,5 \cdot 1,2} = 900\ kgf$

b) $P_{H_2O} = v \cdot e \cdot g = 1 \cdot 1,2 \cdot 0,5 \cdot 1000 \cdot 10 = 6000\ N = 600\ kgf$

$P = \frac{P_{H_2O}}{A_B} = \frac{600}{0,5 \cdot 1,2} = 1000\ kgf$

$P_T = 1000 + 900 = 1900\ kgf$

(8) $P = e \cdot g \cdot H$

ÁGUA: $p = 1000 \cdot 10 \cdot 3 = 30000\ N/m^2$

$1\ N = 0,1\ kgf$

$1\ m^2 = 10000\ cm^2$

$P = \frac{30000 \cdot 0,1}{10000} = 0,3\ kgf/cm^2$

Gasolina: $p = 750 \cdot 10 \cdot 3 = 22500\ N/m^2$

$P = \frac{22500}{10000} \cdot 0,1 = 0,225\ kgf/cm^2$

(9) $P = 4{,}2\ kgf/m^2 = 420000\ N/m^2$

(1)

a) $P = e \cdot g \cdot H \Rightarrow 420000 = 800 \cdot 10 \cdot H$

$H = 52{,}5\ m$

b) $420000 = 1000 \cdot 10 \cdot H \Rightarrow H = 42\ m \Rightarrow$ 42 mcA

c) $420000 = 13600 \cdot 10 \cdot H \Rightarrow H = 30{,}88\ m \Rightarrow$ 3088 mm H_2O

(10)

a) $P = 0{,}28\ kgf/m^2 = 28000\ N/m^2$

$P = e \cdot g \cdot (H+2) = 28000 = 1000 \cdot 10 \cdot (H+2) \Rightarrow H = 0{,}8\ m$

b) $P = 1000 \cdot 10 \cdot (H+5) = 10000 \cdot 5{,}8 = 58000\ N/m^2$

$P = 0{,}58\ kgf/m^2$

(11) $\Delta P = e \cdot g \cdot \Delta H$

$P_1 = e \cdot g \cdot H_1 \Rightarrow P_1 = 13600 \cdot 10 \cdot 0{,}735 = 99960\ N/m^2$

$P_2 = 13600 \cdot 10 \cdot 0{,}59 = 80240\ N/m^2$

$\Delta P = 99960 - 80240 = 19720\ N/m^2$

$\Delta P = e \cdot g \cdot \Delta H$

$19720 = 1{,}125 \cdot 10 \cdot \Delta H$

$\Delta H = 1753\ m$

(12)

$$P = P_0 \cdot e^{-\frac{M}{RT} \cdot gH}$$

$$P = 1 \cdot e^{-\frac{0,0288}{8,314 \cdot 288} \cdot 9,806 \cdot H}$$

$$P = e^{-1,179456 \cdot 10^{-4} \cdot H}$$

- $H = 1000\ m \Rightarrow P = 0,89\ ATm$
- $H = 5000\ m \Rightarrow P = 0,55\ ATm$
- $H = 10000\ m \Rightarrow P = 0,31\ ATm$

(13) p/ $H < 11000\ m$

$$T = 15,04 - 0,00649\,H$$

$$P = 101,29 \cdot \left[\frac{T + 273,1}{288,08}\right]^{5,256}$$

- $H = 1000\ m$ { $T = 8,55°C$, $P = 0,8996\ ATm$
- $H = 5000\ m$ { $T = -17,41°C$, $P = 0,5411\ ATm$
- $H = 10000\ m$ { $T = -49,86°C$, $P = 0,2652\ ATm$

(14) Modelo Atmosfera Padrão 1976

Utilizei a calculadora no site

www.digitaldutch.com/atmoscalc/